

7.19 Use loop equations to determine the velocity and acceleration of point A on link 4. The angular velocity of link 2 is constant at 10 rad/s counterclockwise. Assume that $A^*B^* = 10$ in, $\theta_2 = 60^\circ$, and $\theta_3 = 90^\circ$.

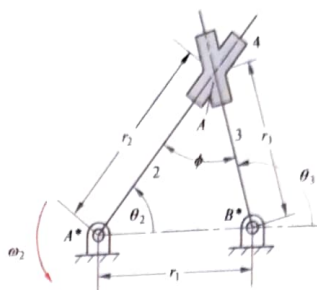


FIGURE P7.19 Linkage for Problem 7.19.

7.20 The oscillating fan shown in Figure P7.20 is to be analyzed as a double rocker. The fan housing is link 2, the motor shaft is connected to link 3, and link 4 is connected from the coupler to the frame. The actual input of the mechanism is the coupler, and ${}^2\omega_3$ is a constant 956 (rad/s) in the counterclockwise direction. Compute the angular velocity and angular acceleration of link 2 if $\theta = 120^\circ$, $A^*B^* = 0.75$ in, $A^*A = B^*B = 3.0$ in, $AB = 0.50$ in.

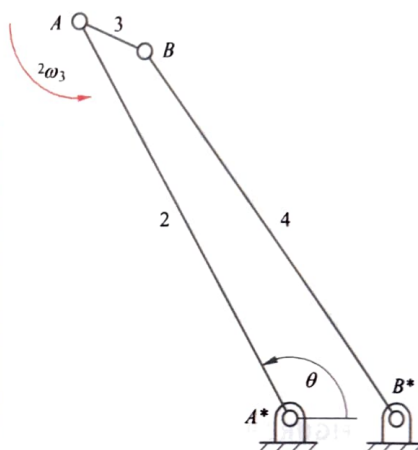


FIGURE P7.20 Fan linkage for Problem 7.20.

7.21 The rear motorcycle suspension can be analyzed as an inverted slider-crank mechanism. The frame of the motorcycle is link 1 and the tire assembly is attached to link 2 at point C. The shock absorber is made up of links 3 and 4. As the bicycle goes over a bump in the position shown, the angular velocity of link 2 relative to the frame, ω_2 , is 5 rad/s CW, and the angular acceleration, α_2 , is 45 rad/s² CW. Compute the angular velocity and angular acceleration of link 3 for the position defined by $\theta = 187^\circ$.

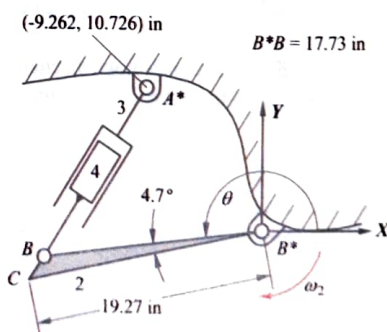


FIGURE P7.21 Motorcycle linkage for Problem 7.21.

7.22 The door-closing linkage shown in Figure P7.22 is to be analyzed as an inverted slider-crank linkage. Link 2 is the door, and links 3 and 4 are the two links of the door closer. Assume that the angular

velocity of the door (link 2) is a constant 3.71 radians per second CW. Compute the angular velocity and angular acceleration of link 4 if the coordinates of B^* are (4.0, -3.5) in.

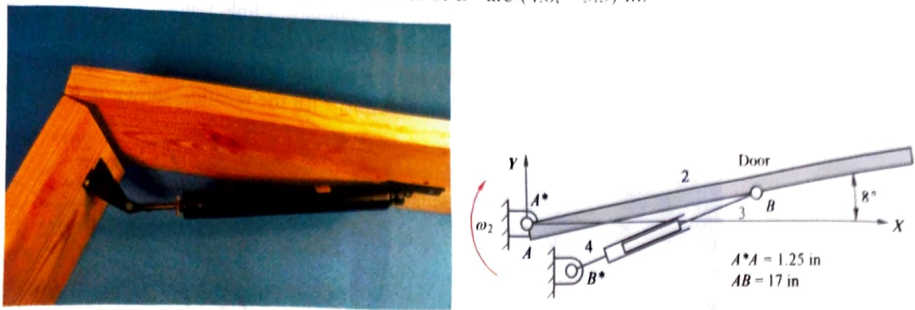


FIGURE P7.22 Door closer linkage for Problem 7.22.

7.23 The general action of a person who is doing pushups can be modeled as a four-bar linkage as shown in Figure P7.23. The floor is the base link and link 4 is the back and legs. Link 2 is the forearm and link 3 is the upper arm. For the purposes of analysis, the motion that is controlled is the motion of link 3 relative to link 2 (elbow joint). Assume that ${}^2\omega_3$ is a constant 3.0 rad/s CCW. Compute the angular velocity and angular acceleration of link 4 if link 2 is oriented at 45° to the horizontal.

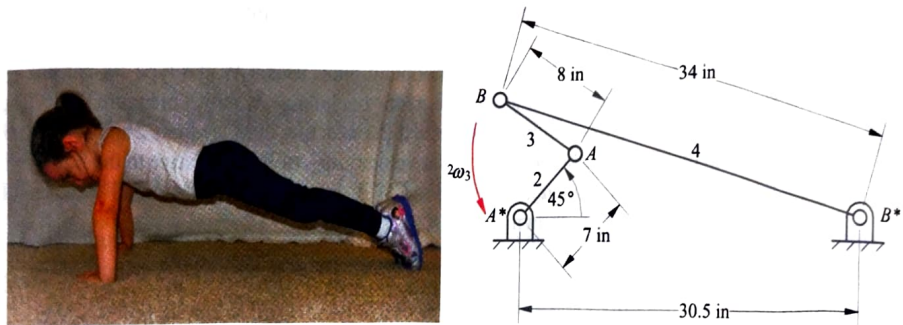


FIGURE P7.23 Linkage representation of push-ups for Problem 7.23.

7.24 A carousel mechanism can be modeled as an inverted slider-crank mechanism as shown in Figure P7.24. Point B is the location of the saddle on the horse. Assume that the angular velocity of the driver (link 2) is a constant 2 rad/s counterclockwise. Compute the velocity and acceleration of B_3 in the position shown if $A^*A = 8.0$ in, $A^*B^* = 101$ in, and $AB = 54$ in.

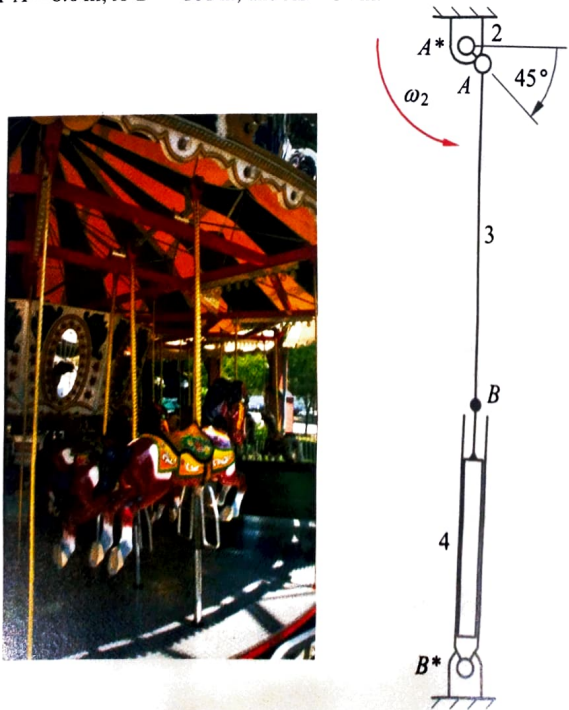


FIGURE P7.24 Carousel linkage for Problem 7.24.

7.25 The shock absorber mechanism on a mountain bicycle is often based on a four-bar linkage as shown in Figure P7.25. When a bump is hit, a resisting torque is applied to link 2 through a shock absorber. The frame of the bike is link 1, the wheel assembly comprises link 3, and the connecting linkage comprises links 2 and 4. As the bicycle goes over a bump in the position shown, the angular velocity, ω_2 , of link 2 relative to the frame is 205 rad/s CW, and the angular acceleration, α_2 , is 60 rad/s² CW. Compute the angular velocity and angular acceleration of link 3 for the position shown in Figure P7.25.

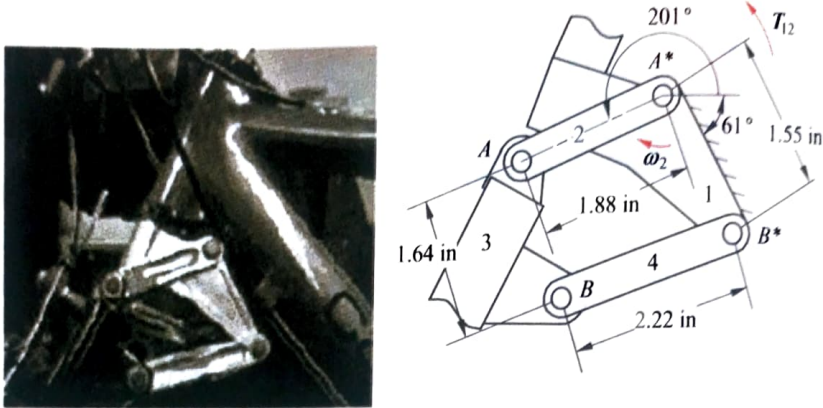


FIGURE P7.25 Shock absorber linkage for Problem 7.25.

7.26 The rear suspension of many mountain bikes uses a Watt's type six-bar linkage where the main support unit containing the rear wheel is a four-bar linkage as shown in Figure P7.26. The frame of the bike is link 1, the wheel assembly is link 3, and the connecting links are links 2 and 4. As the bicycle goes over a bump in the position shown, the angular velocity, ω_2 , of link 2 relative to the frame is 100 rad/s CW, and the angular acceleration, α_2 , is 100 rad/s² CW. Compute the angular velocity and angular acceleration of link 4 for the position shown.

$AB = 15.29$ in	$AA^* = 14.00$ in	$BC = 0.91$ in	$B^*C = 1.92$ in
$B^*C^* = 9.56$ in	$A^*B^* = 6.63$ in	$BB^* = 2.82$ in	

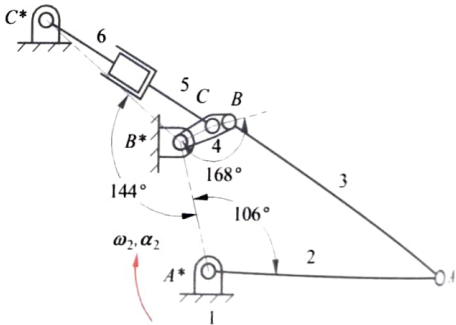


FIGURE P7.26 Bike linkage for Problem 7.26.